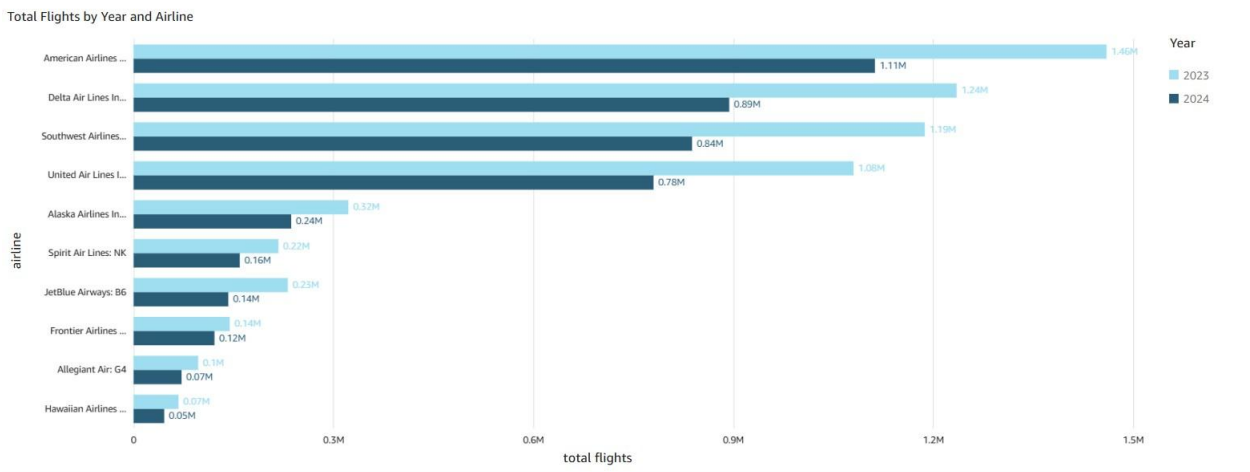


SUMMARY

This report is aimed at understanding flight behaviors and patterns from 2023 to 2024. American Airlines Inc. was found to be the airline with the most flights both in 2023 and 2024. February is the best month to travel in (to avoid delays). The best time to travel in or depart from, would be early morning from 5 AM to 7 AM. Tuesday is the ideal day of the week to travel considering delays as a primary reason for avoiding days like Sunday and Friday. Through the data available in the dataset and from the FAA website (Aircraft Inquiry, n.d.), it was found that there is no evidence which supports the statement “Older planes suffer more delays”. In fact, a small negative relationship was linked to the same. Location based trends revealed cities like Denver and Dallas have more people flying in and cities like New York continue to have the most people coming over the past 2 years. Hagerstown Regional-Richard A. Henson Field of Hagerstown, Maryland was found to be the airport with the highest percentage of flights delayed due to weather conditions. The below expands on these core points and contains visualizations to compare other factors.

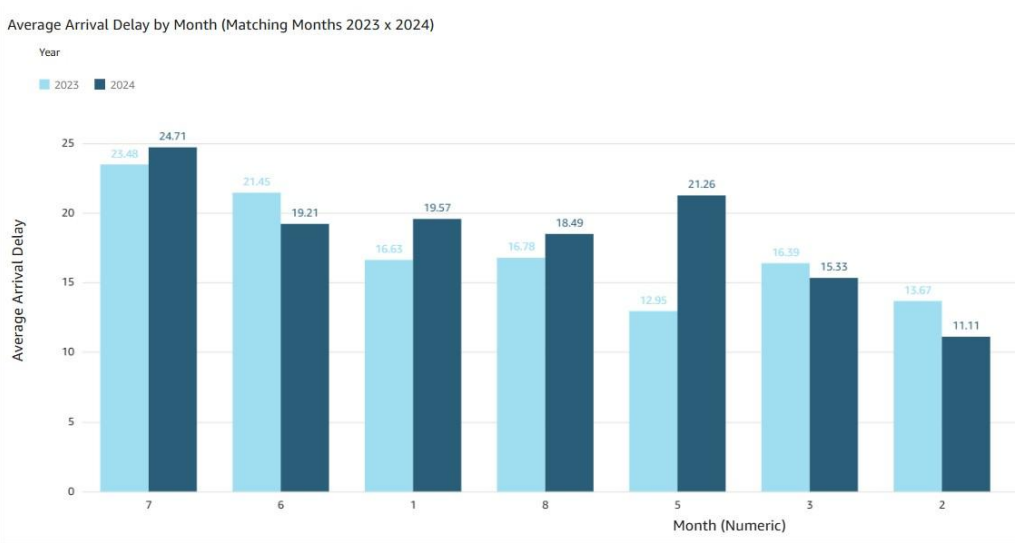
ANALYSIS:

Considering the dataset which contains flight information from 2023 to 2024 and a total of 10445962 flights, American Airlines Inc. was found to be the airline with the most flights both in 2023 and 2024 (1460192 and 1112565 flights respectively) followed by Delta



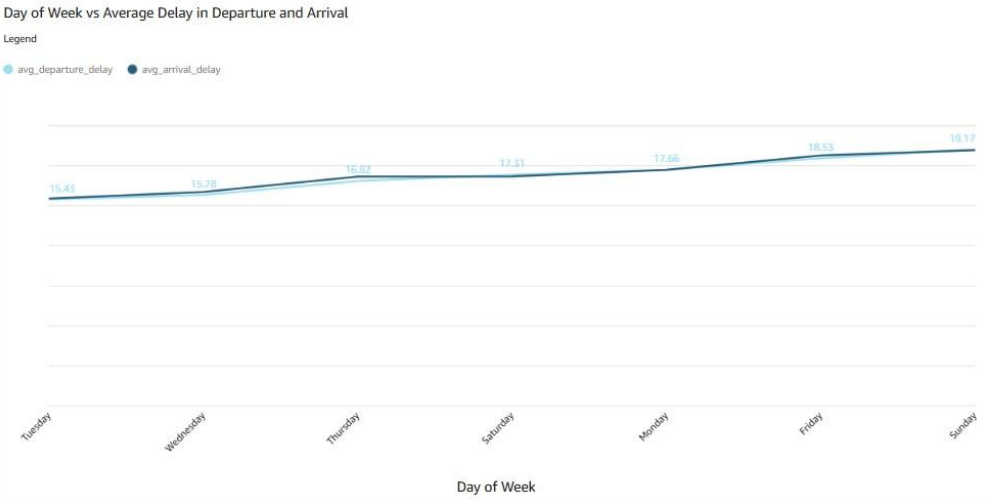
Airlines Inc. with 1235293 and 893986 flights respectively. The above chart depicts the breakdown of the ten airline companies which were part of the dataset and operating in this time interval. The 2024 data indicates an obvious dip mainly because the dataset contains data only for a few months of 2024 compared to 2023 as the calendar year is still progressing. However, from the above bar diagram, we can see that the airline companies have performed at a similar level through these couple of years and have continued operating similar number of flights proportionally.

Certain factors like weather and security could cause delays of flights, and the data during these months suggest that the best month to travel and minimize



delays would be February followed by March. For the months where data is present from 2023 and 2024, the average departure and arrival delay was found to be the least in February and March, with July having the most delays (both on departure and arrival side delays). May 2023 was the month with the least average delays, however in 2024 the average delays were second highest which put the month of May out of contention as the best month to travel in. December 2023 had the least average delay time with an average arrival delay of 11.06 minutes.

The best day to travel on was found to be Tuesday with the average arrival and departure delay being 15.52 and 15.43 minutes respectively. With average delay times of 19.17 and 18.53 minutes, Sunday and Friday respectively appear to be the worst days to travel in, if only delay time was in consideration.



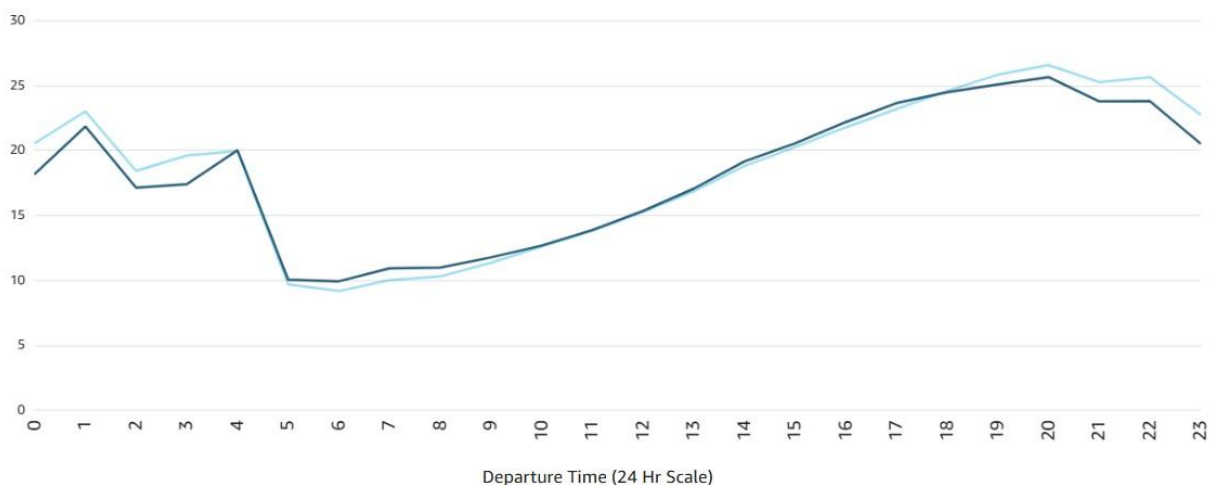
From the graph on the right, the best days to travel in would be Tuesday, Wednesday and Thursday.

The best hours to travel as a passenger would be the early morning hours (considering only delays as a factor). The graph depicts a 24-hour scale of time on the horizontal line, and it shows how gradually the average arrival and departure delay rises to 23.77 and 25.61 minutes respectively before dipping again to the lows of sub-10-minute average delay. We can conclude from the graphical representation that the best hours to travel in (or departure time) would encompass an interval between 5 AM and 7 AM. This interval of time would be when a passenger can expect the least average delays.

Departure Time vs Average Departure & Arrival Delay

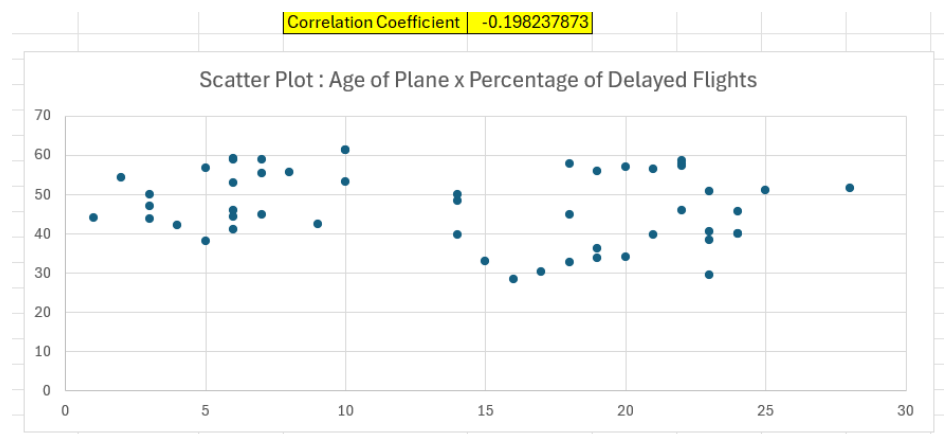
Legend

● avg_departure_delay ● avg_arrival_delay



As the dataset contains data about flights, it is possible to investigate the details of the aircraft itself to determine some relationships. Using the FAA Aircraft Inquiry page (Aircraft Inquiry, n.d.), the age of flight could be tracked using the tail number associated with the flight. With the

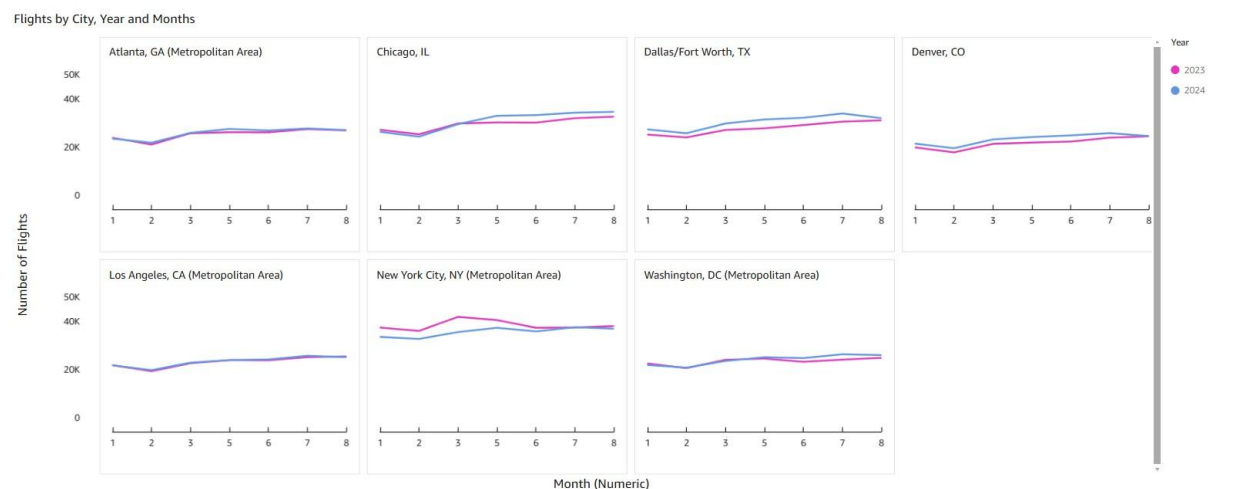
hypothesis that older flights tend to have more delays, two kinds of tests were performed aka a T-Test and a correlation test. The figure on the right displays the age of the aircraft on the



horizontal line and percentage of delayed flights on the vertical. The 'scatter' plot shows a very minor relationship between age of flight and the chances of it getting delayed. This

was supported by the correlation coefficient obtained by a Pearson Correlation test (which tests the linear relation between two sets of data) which was obtained as -0.198, which signifies a very weak negative relationship. This can be observed in the scatter plot as well. To expand on this, a T Test was performed with the hypothesis (H0)/Null Hypothesis that “The age of a plane does not significantly affect flight delays” and the Alternative Hypothesis (H1) that “The age of a plane significantly affects flight delays”. The p-value **T-TEST: p - value 0.411622** (probability value) obtained from the t-test was 0.41, which is much greater than the significance threshold ($\alpha=0.05$). This indicates that the difference in average flight delays between older and newer planes is not statistically significant. The result implies that delays are likely influenced by other factors, such as operational practices, weather conditions, or route-specific characteristics, rather than the age of the aircraft. (NB: The T-Test was conducted as a Two-Tailed and Two-Sample Unequal Variance).

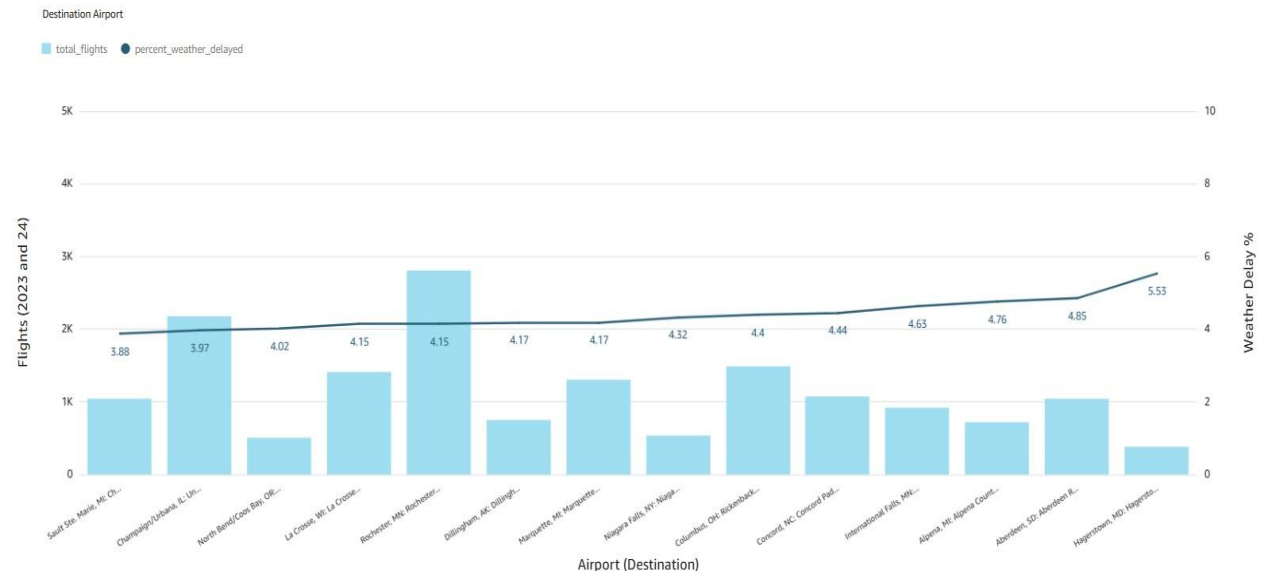
Looking at patterns of people flying to and from different cities, looking at flight data going into several key cities would give insights into some patterns. The top 7 cities (with respect to number of flights flying in/arriving) from 2023 and 2024 were investigated and the trends can be observed below. New York City (Metropolitan Area) was the city with the largest



number of flights coming in with 392600 in 2023 and 260670 already in 2024. The graph above shows only those months where data was available in the previous year so that the comparison could be made seamlessly. New York City (Metropolitan Area) despite a dip in the earlier months of 2023 seems to have recovered in the latter half of 2024. Cities like Chicago, Dallas and Denver have seen a noticeable increase in the frequency of flights coming in 2024 which could be a sign of more people coming in. On the other hand, Los Angeles and Atlanta have remained almost the same between 2023 and 2024.

Weather delays affect flight landings and ultimately flight delays. The below are the airports

Weather Delay Percentage by Total Flights to the Airport



where the weather affects landing and ultimately causes delay. Hagerstown Regional-Richard A. Henson Field of Hagerstown, Maryland is the airport with the highest percentage of delays due to weather conditions. Over the span of the entire dataset of 10 million plus flights, it had a 5.53% of all flights delayed due to weather. Rochester International of Rochester, Minnesota with a higher frequency of flights also stands out with a 4.15% of all flights delayed due to weather.

References

Aircraft Inquiry. (n.d.). Retrieved December 4, 2024, from Federal Aviation Administration (.gov): <https://registry.faa.gov/aircraftinquiry/search/nnumberinquiry>

Appendix

Steps taken for transformation:

- 1) Load in all lookup tables via Glue Crawler using a Visual ETL file on to same database schema.
- 2) Analyze 2023 and 2024 csv's and documentation csv to determine columns to keep in a combined dataset.

- The screenshot displays the AWS Glue console interface. At the top, the navigation bar includes the AWS logo, a search bar, and the user's name 'N. Virginia'. Below this, the breadcrumb trail shows 'AWS Glue > Jobs'. A 'Create example job' button is located in the top right corner. The main heading is 'Example jobs info'. Underneath, the 'Your jobs (11) info' section features a search bar and a table of jobs. The table has columns for 'Job name', 'Type', 'Created by', 'Last modified', and 'AWS Glue version'. The jobs listed are: flights_2023_2, flights_2023_1, flag_indicator_ETL, unique_carriers_ETL, city_market_ETL, Carrier_History_ETL, Cancellation_ETL, Airport_Sequence_ETL, Airport_ETL, and Airline_ETL. The 'Airline_ETL' job is highlighted in blue. To the right of the table, there are 'Actions' and 'Run job' buttons. A 'Unsaved job found' notification is visible at the bottom right, stating 'We found an unsaved job, do you wish to restore it?'. The footer includes 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates.

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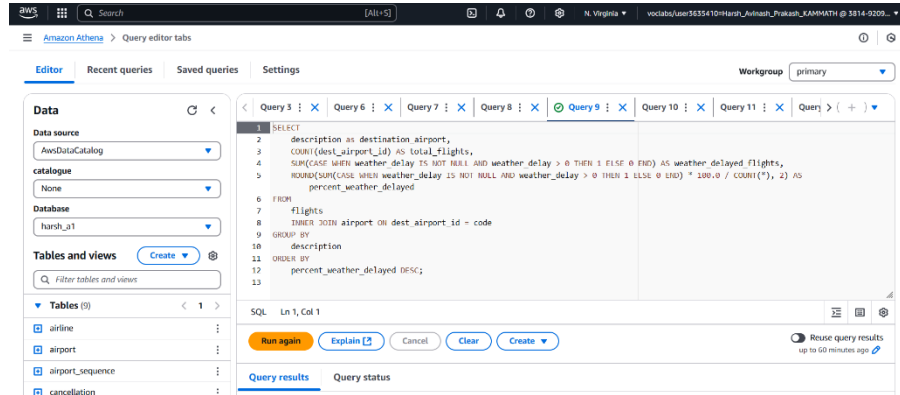
chosen to dump data into the schema was chosen wrongly and identified only after upload of July 2023 dataset.

- 7) Following this approach all data was dumped into the schema and accessed via Athena.

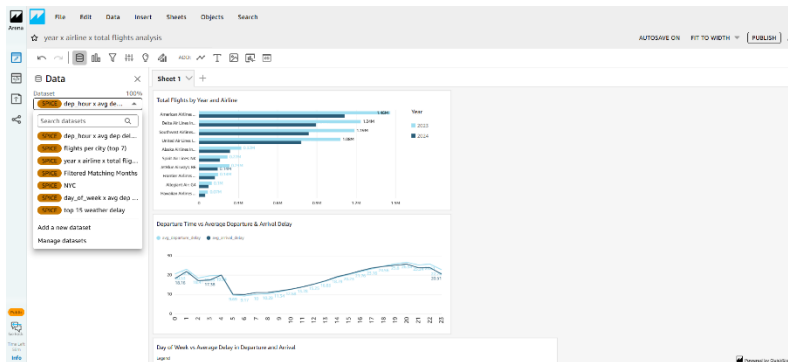
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- The screenshot displays the Databricks workspace interface. At the top, there's a navigation bar with the Databricks logo, a search bar, and user information (N. Virginia). Below this, the breadcrumb path is 'workspace / user3635410 / home / Analysis / Projects / KAMMATH / @ 5814-9205'. The main header shows the workflow name 'flights_2023_2' and its last modified time '12/1/2023, 4:53:54 PM'. There are buttons for 'Actions', 'Save', and 'Run'. The 'Visual' tab is selected, showing a workflow diagram with five steps: 'Data source - D3 dataset', 'Transform - OpenAI Text to SQL', 'Transform - OpenAI Text to JSON', 'Transform - OpenAI Text to CSV', and 'Data target - D3 dataset'. The workflow is currently in a 'preview' state, as indicated by the 'Upgrade analysis - preview' button. A sidebar on the right contains icons for various workspace functions.

table which contains 27 columns (a few columns were removed from the original dataset as they were not required as part of analysis). Queries were constructed on Athena and output files were collected through download option available and the output was obtained as csv files which were used for further analysis.

9) Concepts like group by, order by, aggregation and joins were used to formulate queries for analysis.



10) For plotting graphs and visualizations, Quicksight was used to generate them using



the obtained result csv's.

11) For certain mathematical operations, Excel was used to perform Correlation analysis and T Test for Hypothesis Testing. From the result set which contained information regarding 6000+ unique planes, a random sample of 51 planes

were chosen and the age of each of these planes obtained from the FAA website. Using this data, correlation was performed on percentage chance of delay with age of aircraft and hypothesis testing conducted on average delay time with age of aircraft.

